

Intermediate Elective

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Packages

```
# general use and cleaning
library(tidyverse)
library(here)
library(janitor)
library(snakecase)
library(lubridate)
library(dplyr)

# wrapping axis labels
library(scales)

# reading in .xlsx files
library(readxl)

# visualizing missing data
library(naniar)

# arranging plots
library(patchwork)
```

Data

```
# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))
```

```
# aquatic invertebrates
aq_ins <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Aquatic Insects")

# site information
sites <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "sites")

# water quality information
water_quality <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Water Quality",
  na = c("", "n/a", "N/A", "over", "173+"))

# creating new object from sites
ncos_sites <- sites |>
  # filter to only include NCOS sites sampled four times a year
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")
```

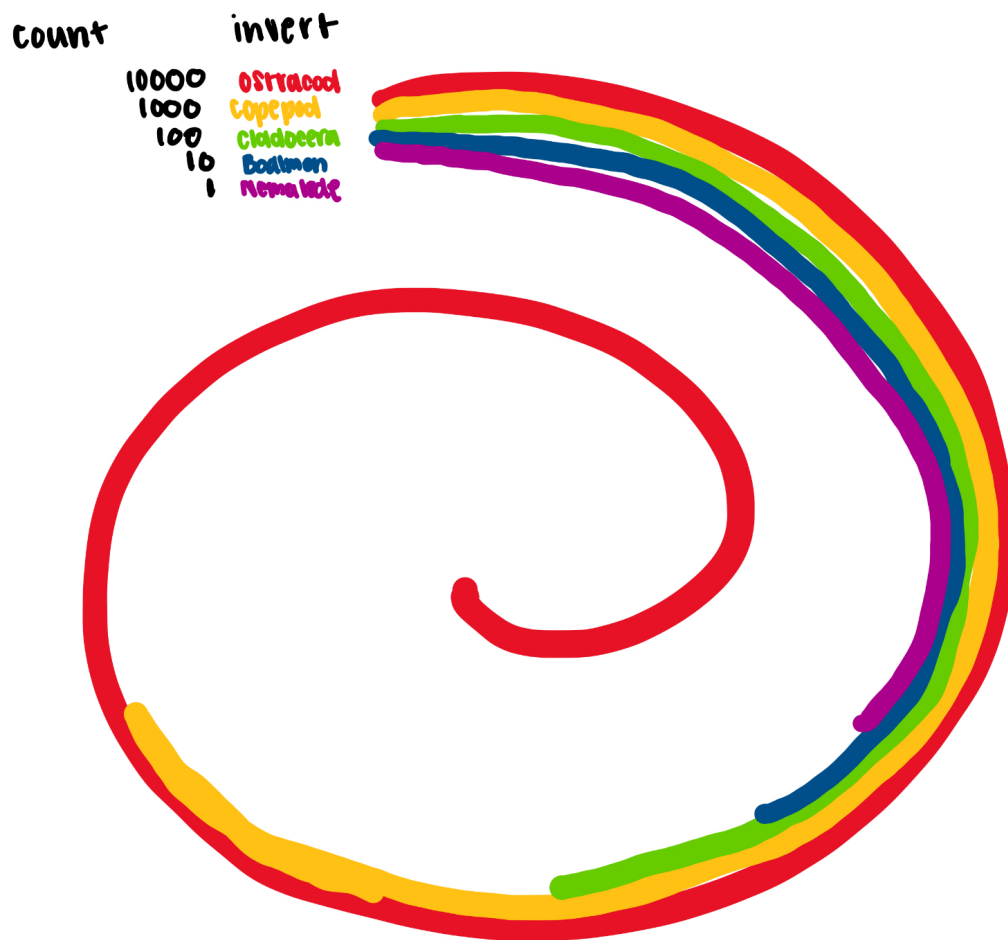
1. Explain your inspiration

- I picked the spiral looking graphic labeled 2021_05_wealth_income graphic by Ijeamaka Anyenes.
- This graphic makes sense for my dataset of choice because it is able to display differences in abundance in a fun yet clear and simple way.
- The visualization will show the 5 most abundant inverts during 2024. Each line will represent a different invert and have its own color while the length of the line represents abundance.

2. Plan your figure

- Final visual goal: compare the abundances of the most common inverts found at ncos in 2024

```
knitr:: include_graphics(here("code", "IMG_2970.jpg"))
```



#3. Code your figure

Create data frame

```
invert_2024 <- aq_ins |>
  clean_names() |>
  rename(date = date_on_vial) |>
  filter(year(date) == 2024) |>
  select(ostracod,
         copepod,
         hemiptera_corixidae_boatman,
         cladocera,
```

```

      nematode) |>
pivot_longer(cols = ostracod:nematode,
             names_to = "taxon",
             values_to = "abundance") |>
mutate(
  abundance = replace_na(abundance, 0),
  taxon = str_to_title(str_replace_all(taxon, "_", " "))
) |>
group_by(taxon) |>
summarise(total_abundance = sum(abundance), .groups = "drop")

```

```

ring_data <- invert_2024 |>
  arrange(desc(total_abundance)) |>
  mutate(
    abundance_pct = total_abundance / sum(total_abundance),
    pct_label = paste0(round(abundance_pct * 100, 1), "%"),
    label = paste(taxon, pct_label),
    ring = row_number(),
    inner = ring,
    outer = ring + 0.45,
    start = 0,
    end = abundance_pct * 100
  )

```

```

final_plot <- ggplot(ring_data) +
  geom_rect(
    aes(
      xmin = start,
      xmax = end,
      ymin = inner,
      ymax = outer,
      group = taxon
    ),
    fill = "#0c1931"
  ) +
  geom_rect(
    aes(
      xmin = 0,
      xmax = 100,
      ymin = inner,
      ymax = outer,
      group = taxon
    )
  )

```

```

    ),
    fill = NA,
    color = "#cdcdc0",
    linewidth = 2
  ) +
  geom_text(
    aes(
      x = 0,
      y = outer + 0.2,
      label = label
    ),
    hjust = 0,
    size = 3,
    fontface = "bold"
  ) +
  coord_polar(theta = "x", start = pi / 2) +
  scale_x_continuous(limits = c(0, 100), expand = c(0, 0)) +
  scale_y_continuous(limits = c(0, max(ring_data$outer) + 1)) +
  labs(title = "Aquatic Invertebrate\nAbundance in 2024") +
  theme_void() +
  theme(
    plot.title = element_text(hjust = 0.5, size = 20, face = "bold"),
    panel.background = element_rect(fill = "#f4f2f2", color = NA),
    plot.background = element_rect(fill = "#f4f2f2", color = NA),
    legend.position = "none"
  )
)

final_plot

```

Aquatic Invertebrate Abundance in 2024

